



自动控制

自动控制原理

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第四章 频域分析法



提 纲

● 频域分析法

- 频率特性的基本概念
- 幅相频率特性 (Nyquist图)
- 对数频率特性 (Bode图)
- 频域稳定判据
- 闭环系统的频率特性
- 系统校正



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频率特性的基本概念

● 频域分析法的特点

- 研究稳态正弦响应的幅值和相角随频率的变化规律
- 由开环频率特性研究闭环稳定性及性能
- 图解分析法，方便，实用
- 有一定的近似性
- 工程上广泛采用

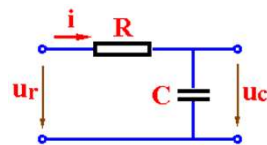


频率特性的基本概念

● 频率响应

- 线性系统稳态正弦响应的幅值、相角随输入频率变化的规律性

例：RC 电路如图所示， $u_r(t) = A \sin \omega t$ ，求 $u_c(t) = ?$



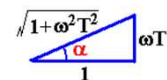
$$G(s) = \frac{U_c(s)}{U_r(s)} = \frac{1}{CRs + 1} \stackrel{T=CR}{=} \frac{1}{Ts + 1} = \frac{1/T}{s + 1/T}$$

$$U_c(s) = \frac{1/T}{s + 1/T} \cdot \frac{A\omega}{s^2 + \omega^2} = \frac{C_0}{s + 1/T} + \frac{C_1 s + C_2}{s^2 + \omega^2}$$

$$U_c(s) = \frac{A\omega T}{1 + \omega^2 T^2} \cdot \frac{1}{s + 1/T} + \frac{A}{\sqrt{1 + \omega^2 T^2}} \left[\frac{1}{\sqrt{1 + \omega^2 T^2}} \cdot \frac{\omega}{s^2 + \omega^2} - \frac{T\omega}{\sqrt{1 + \omega^2 T^2}} \cdot \frac{s}{s^2 + \omega^2} \right]$$

$$u_c(t) = \frac{A\omega T}{1 + \omega^2 T^2} e^{-\frac{t}{T}} +$$

$$= \frac{A\omega T}{1 + \omega^2 T^2} e^{-\frac{t}{T}} + \frac{A}{\sqrt{1 + \omega^2 T^2}} \sin(\omega t - \arctan \omega T)$$





频率特性的基本概念

● 频率特性

$$r(t) = A \sin \omega t \rightarrow \boxed{\frac{1}{Ts+1}} \rightarrow c_s(t) = \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t - \arctan \omega T)$$

- 对于稳定的线性定常系统，输入一个正弦信号，其稳态响应也是一个同频率的正弦信号
- 稳态响应的幅值与输入正弦信号幅值之比记为 $|G(j\omega)|$ ，相位之差（相移）记为 $\phi = \angle G(j\omega)$
- $G(j\omega) = |G(j\omega)| \angle G(j\omega)$ ：表示稳定线性定常系统的稳态输出和输入正弦波之间的关系
- 故称 $G(j\omega)$ 为频率响应函数、或系统的频率特性



$$r(t) = A \sin \omega t \rightarrow \boxed{\frac{1}{Ts+1}} \rightarrow c_s(t) = \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t - \arctan \omega T)$$

● 频率特性 $G(j\omega)$ 的定义

- 定义一： $G(j\omega) = |G(j\omega)| \angle G(j\omega)$

$$\begin{cases} |G(j\omega)| = \frac{|c_s(t)|}{|r(t)|} = \frac{1}{\sqrt{1+\omega^2 T^2}} \\ \angle G(j\omega) = \angle c_s(t) - \angle r(t) = -\arctan \omega T \end{cases}$$

- 定义二： $G(j\omega) = G(s)|_{s=j\omega}$

$$\frac{1}{\sqrt{1+\omega^2 T^2}} \angle -\arctan \omega T = \left| \frac{1}{1+j\omega T} \right| \angle \frac{1}{1+j\omega T} = \frac{1}{1+j\omega T} = \frac{1}{Ts+1} \Big|_{s=j\omega}$$

- 定义三： $G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$

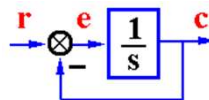
$$G(s) = C(s)/R(s) \quad s = j\omega$$



频率特性的基本概念

● 频率特性 $G(j\omega)$ 的定义

例：系统结构图如图所示， $r(t) = 3\sin(2t + 30^\circ)$ ，求 $c_s(t)$, $e_s(t)$



$$\text{解. } \Phi(s) = \frac{1}{s+1} \begin{cases} |\Phi(j\omega)| = \left| \frac{1}{1+j\omega} \right| = \frac{1}{\sqrt{1+\omega^2}} \stackrel{\omega=2}{=} \frac{1}{\sqrt{5}} = \frac{|c_s(t)|}{3} \\ \angle \Phi(j\omega) = -\arctan \omega \stackrel{\omega=2}{=} -63.4^\circ = \angle c_s(t) - \angle r(t) = \angle c_s(t) - 30^\circ \end{cases}$$

$$\begin{cases} |c_s(t)| = 3/\sqrt{5} \\ \angle c_s(t) = -63.4^\circ + 30^\circ = -33.4^\circ \end{cases} \quad c_s(t) = \frac{3}{\sqrt{5}} \sin(2t - 33.4^\circ)$$

$$\Phi_e(s) = \frac{s}{s+1} \begin{cases} |\Phi_e(j\omega)| = \left| \frac{j\omega}{1+j\omega} \right| = \frac{\omega}{\sqrt{1+\omega^2}} \stackrel{\omega=2}{=} \frac{2}{\sqrt{5}} = \frac{|e_s(t)|}{3} \\ \angle \Phi_e(j\omega) = 90^\circ - \arctan \omega \stackrel{\omega=2}{=} 90^\circ - 63.4^\circ = \angle e_s(t) - 30^\circ \end{cases}$$

$$\begin{cases} |e_s(t)| = 6/\sqrt{5} \\ \angle e_s(t) = 26.6^\circ + 30^\circ = 56.6^\circ \end{cases} \quad e_s(t) = \frac{6}{\sqrt{5}} \sin(2t + 56.6^\circ)$$



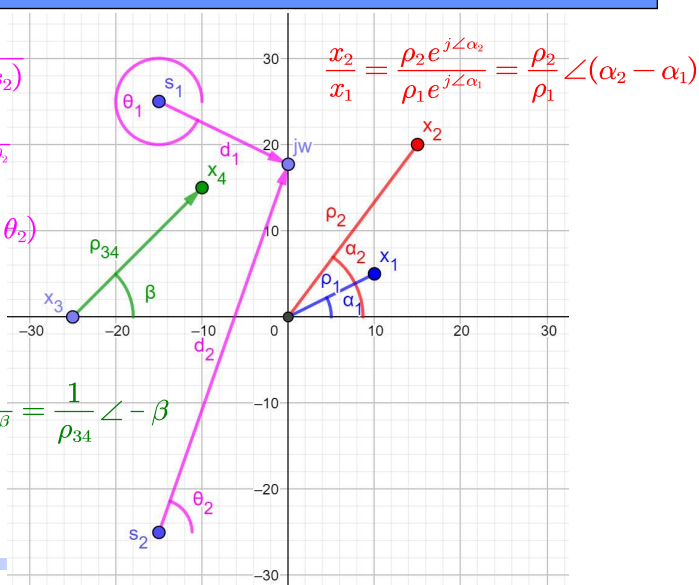
频率特性的基本概念

$$\frac{1}{(j\omega - s_1)(j\omega - s_2)}$$

$$= \frac{1}{d_1 e^{j\angle\theta_1} \cdot d_2 e^{j\angle\theta_2}}$$

$$= \frac{1}{d_1 d_2} \angle -(\theta_1 + \theta_2)$$

$$\frac{1}{x_4 - x_3} = \frac{1}{\rho_{34} e^{j\angle\beta}} = \frac{1}{\rho_{34}} \angle -\beta$$

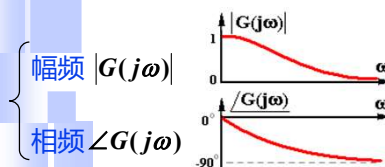




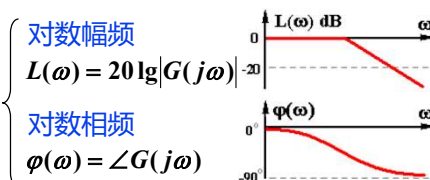
频率特性的基本概念

● 频率特性 $G(j\omega)$ 的表示方法

I. 频率特性



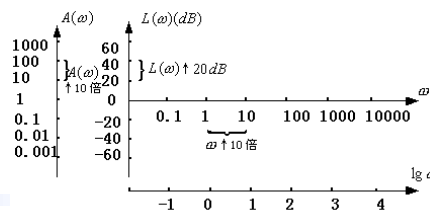
II. 对数频率特性(Bode)



$G(j\omega)$ 及 ω 的变化范围太大, 故用对数坐标表示

- 纵坐标以幅值的对数分贝数为刻度, 均匀分布
- 横坐标按频率对数 $\lg \omega$ 标尺刻度, 但标出的是实际的 ω 值, 不均匀分布

——这种坐标系称为**半对数坐标系**



频率特性的基本概念

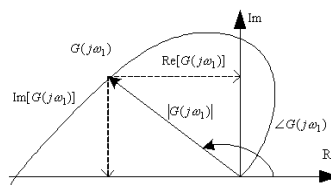
● 频率特性 $G(j\omega)$ 的表示方法

III. 幅相特性(Nyquist)

● 极坐标图

$G(j\omega)$ 是复数, 亦可看作一个矢量

ω 从 $0 \rightarrow \infty$ 变化时, 矢量 $G(j\omega)$ 的端点在复平面上的运动轨迹



- 正相角按正实轴方向反时针旋转定义
- 用来表示频率特性 $G(j\omega)$ 的平面称为**G平面**
- $G(j\omega)$ 的幅相特性是s平面上的虚轴通过 $G(j\omega)$ 在G平面上的映射



频率特性的基本概念

● 频率特性 $G(j\omega)$ 的表示方法

III. 幅相特性(Nyquist)

- $G(j\omega)$ 是 s 平面上的虚轴通过 $G(j\omega)$ 在 G 平面上的映射

$$G(j\omega) = G(s)|_{s=j\omega}$$

$$G(s) = \frac{1}{0.5s+1} = \frac{2}{s+2}$$

$$G(j\omega) = \frac{2}{2+j\omega}$$



频率特性的基本概念

● 频率特性 $G(j\omega)$ 的表示方法

III. 幅相特性(Nyquist)

- $G(j\omega)$ 是 s 平面上的虚轴通过 $G(j\omega)$ 在 G 平面上的映射

$$G(j\omega) = G(s)|_{s=j\omega}$$

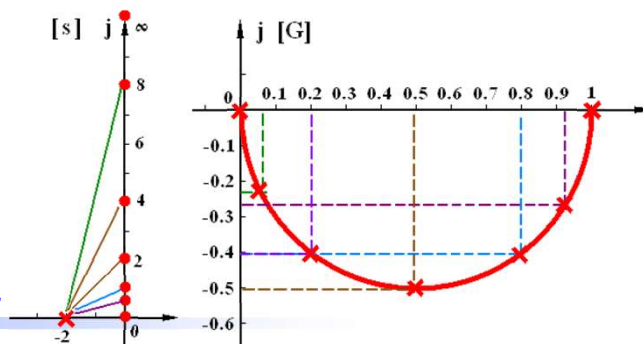
$$G(s) = \frac{1}{0.5s+1} = \frac{2}{s+2}$$

$$G(j\omega) = \frac{2}{2+j\omega}$$

$$G(j\omega) = \frac{2}{(-2) \rightarrow j\omega}$$

$$|G(j\omega)| = \frac{2}{|(-2) \rightarrow j\omega|}$$

$$\angle G(j\omega) = \angle 2 - \angle(-2) \rightarrow j\omega$$





提 纲

● 频域分析法

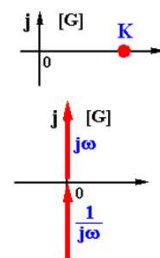
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幅相频率特性

● 典型环节的幅相特性曲线

- (1) 比例环节 $G(s) = K \quad G(j\omega) = K \quad \begin{cases} |G| = K \\ \angle G = 0^\circ \end{cases}$
- (2) 微分环节 $G(s) = s \quad G(j\omega) = j\omega \quad \begin{cases} |G| = \omega \\ \angle G = 90^\circ \end{cases}$
- (3) 积分环节 $G(s) = \frac{1}{s} \quad G(j\omega) = \frac{1}{j\omega} \quad \begin{cases} |G| = 1/\omega \\ \angle G = -90^\circ \end{cases}$
- (4) 惯性环节 $G(s) = \frac{1}{Ts+1} \quad G(j\omega) = \frac{1}{1+j\omega T} \quad \begin{cases} |G| = \frac{1}{\sqrt{1+\omega^2 T^2}} \\ \angle G = -\arctan \omega T \end{cases}$





幅相频率特性

(4) 惯性环节 $G(s) = \frac{1}{Ts+1}$ $G(j\omega) = \frac{1}{1+j\omega T}$

证明：惯性环节的幅相特性为半圆

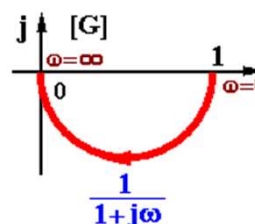
$$G(j\omega) = \frac{1}{1+j\omega T} = \frac{1-j\omega T}{1+\omega^2 T^2} = X + jY$$

$$\Rightarrow \omega T = -\frac{Y}{X}$$

$$X = \frac{1}{1+\omega^2 T^2} = \frac{1}{1+(Y/X)^2}$$

$$\Rightarrow X^2 - X + Y^2 = 0$$

$$\left(X - \frac{1}{2}\right)^2 + Y^2 = \left(\frac{1}{2}\right)^2 \quad \xrightarrow{Y = -\omega T X} \text{(下半圆)}$$



幅相频率特性

(4) 惯性环节 $G(s) = \frac{1}{Ts+1}$ $G(j\omega) = \frac{1}{1+j\omega T}$

例：系统的幅相曲线如图所试，求系统的传递函数。

由曲线形状有 $G(s) = \frac{K}{Ts+1}$

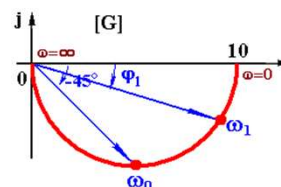
由起点: $G(j0) = \frac{K}{1+j0T} = K = 10 \angle 0^\circ$ $K = 10$

由45度角: $\angle G(j\omega_0) = \angle \frac{K}{1+j\omega_0 T} = -\arctan \omega_0 T = -45^\circ$

$$\omega_0 T = \tan 45^\circ = 1 \quad T = 1/\omega_0$$

由 φ_1 : $\angle G(j\omega_1) = \angle \frac{K}{1+j\omega_1 T} = -\arctan \omega_1 T = -\varphi_1$

$$\omega_1 T = \tan \varphi_1 \quad T = \tan \varphi_1 / \omega_1$$



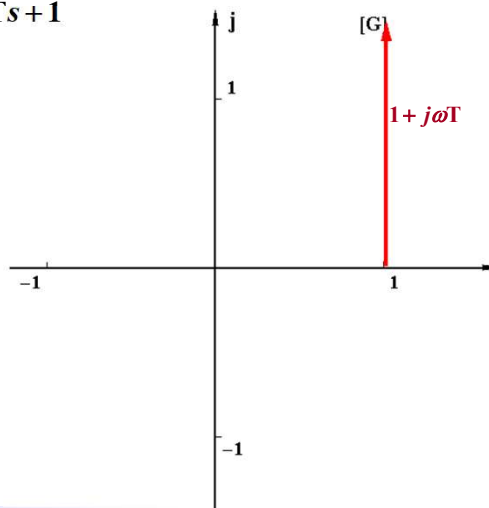


幅相频率特性

(5) 一阶微分环节 $G(s) = Ts + 1$

$$G(j\omega) = 1 + j\omega T$$

$$\begin{cases} |G| = \sqrt{1 + \omega^2 T^2} \\ \angle G = \arctan \omega T \end{cases}$$



幅相频率特性

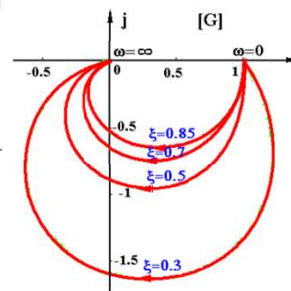
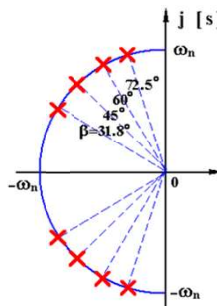
(6) 振荡环节 $G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{1}{(\frac{s}{\omega_n})^2 + 2\xi\frac{s}{\omega_n} + 1} = \frac{\omega_n^2}{(s - \lambda_1)(s - \lambda_2)}$

$$G(j\omega) = \frac{1}{1 - \frac{\omega^2}{\omega_n^2} + j2\xi\frac{\omega}{\omega_n}}$$

$$G(j0) = 1 \angle 0^\circ$$

$$G(j\infty) = 0 \angle -180^\circ$$

$$\begin{cases} |G| = \frac{1}{\sqrt{[1 - \frac{\omega^2}{\omega_n^2}]^2 + [2\xi\frac{\omega}{\omega_n}]^2}} \\ \angle G = -\arctan \frac{2\xi\frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}} \end{cases}$$





幅相频率特性

(6) 振荡环节

● 谐振频率 ω_r 和谐振峰值 M_r

$$|G| = \frac{1}{\sqrt{\left[1 - \frac{\omega^2}{\omega_n^2}\right]^2 + \left[2\xi \frac{\omega}{\omega_n}\right]^2}} = \Delta^{-\frac{1}{2}}$$

$$\frac{d}{d\omega} |G| = 0$$

$$-\frac{1}{2} \Delta^{-\frac{3}{2}} \frac{d}{d\omega} \left\{ \left[1 - \frac{\omega^2}{\omega_n^2}\right]^2 + \left[2\xi \frac{\omega}{\omega_n}\right]^2 \right\} = 0$$

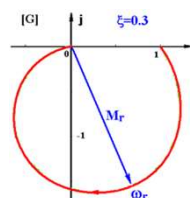
$$-\frac{1}{2} \Delta^{-\frac{3}{2}} \left\{ 2 \left[1 - \frac{\omega^2}{\omega_n^2}\right] \left[-2 \left(\frac{\omega}{\omega_n^2}\right)\right] + 2 \left[2\xi \frac{\omega}{\omega_n}\right] \left(\frac{2\xi}{\omega_n}\right) \right\} = 0$$

$$-\frac{1}{2} \Delta^{-\frac{3}{2}} \frac{4\omega}{\omega_n^2} \left[-1 + \frac{\omega^2}{\omega_n^2} + 2\xi^2\right] = 0 \quad \text{等号能成立吗?}$$

$$\frac{\omega^2}{\omega_n^2} = 1 - 2\xi^2$$

$$\omega_r = \omega_n \sqrt{1 - 2\xi^2}$$

$$M_r = |G(j\omega_r)| = \frac{1}{2\xi \sqrt{1 - \xi^2}}$$



$\xi^2 \leq \frac{1}{2}$ 时能成立, 否则导数恒为负, 幅值随 ω 单调减小



幅相频率特性

(6) 振荡环节

● 谐振频率 ω_r 和谐振峰值 M_r

$$\frac{\omega^2}{\omega_n^2} = 1 - 2\xi^2$$

$$\omega_r = \omega_n \sqrt{1 - 2\xi^2}$$

$$M_r = |G(j\omega_r)| = \frac{1}{2\xi \sqrt{1 - \xi^2}} = \frac{1}{\sin 2\beta}$$

$$\xi > 0.707 \quad (\beta < 45^\circ)$$

$$1 - 2\xi^2 < 0 \quad \omega_r, M_r \text{ 不存在}$$

$$\xi = 0.707 \quad (\beta = 45^\circ)$$

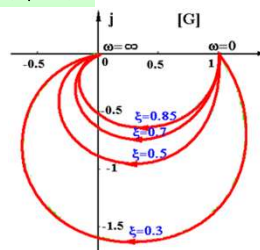
$$1 - 2\xi^2 = 0 \quad \begin{cases} \omega_r = 0 \\ M_r = 1 \end{cases}$$

$$0 < \xi < 0.707 \quad (45^\circ < \beta < 90^\circ)$$

$$1 - 2\xi^2 > 0 \quad \begin{cases} \omega_r = \omega_n \sqrt{1 - 2\xi^2} < \omega_n \\ M_r = \frac{1}{2\xi \sqrt{1 - \xi^2}} > 1 \end{cases}$$

$$\xi = 0 \quad (\beta = 90^\circ)$$

$$1 - 2\xi^2 = 1 \quad \begin{cases} \omega_r = \omega_n \\ M_r = \infty \end{cases}$$





幅相频率特性

(6) 振荡环节

例：系统的幅相曲线如图所示，求传递函数

由曲线形状有 $G(s) = \frac{K}{\frac{s^2}{\omega_n^2} + 2\xi \frac{s}{\omega_n} + 1}$

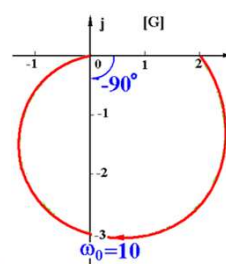
由起点： $G(j0) = K \angle 0^\circ$ $K = 2$

由 $\varphi(\omega_0)$ ： $\angle G(j\omega_0) = -90^\circ$ $\omega_0 = \omega_n = 10$

由 $|G(\omega_0)|$ ： $|G(\omega_0)| = 3 = \frac{\omega_0 = \omega_n}{2\xi} \frac{K}{\omega_n^2} = \frac{2}{2\xi} \frac{2}{10^2}$ $\xi = \frac{1}{3}$

$G(s) = \frac{2 \times 10^2}{s^2 + 2 \times \frac{1}{3} \times 10s + 10^2} = \frac{200}{s^2 + 6.67s + 100}$

$$\begin{cases} |G| = \frac{K}{\sqrt{[1 - \frac{\omega^2}{\omega_n^2}]^2 + [2\xi \frac{\omega}{\omega_n}]^2}} \\ \angle G = -\arctan \frac{2\xi \frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}} \end{cases}$$



幅相频率特性

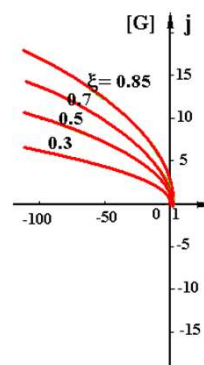
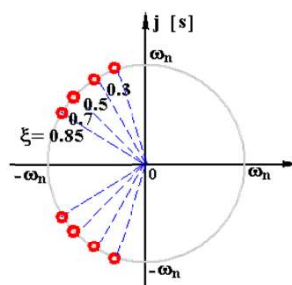
(7) 二阶微分环节

$$G(s) = T^2 s^2 + 2\xi Ts + 1 = \left(\frac{s}{\omega_n}\right)^2 + 2\xi \frac{s}{\omega_n} + 1 = \frac{1}{\omega_n^2} (s - z_1)(s - z_2)$$

$$G(j\omega) = 1 - \frac{\omega^2}{\omega_n^2} + j2\xi \frac{\omega}{\omega_n}$$

$$|G| = \sqrt{[1 - \frac{\omega^2}{\omega_n^2}]^2 + [2\xi \frac{\omega}{\omega_n}]^2}$$

$$\angle G = \arctan \frac{2\xi \frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}$$

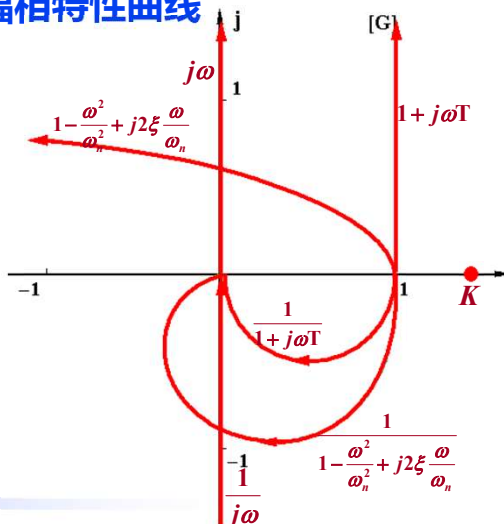




幅相频率特性

● 典型环节的幅相特性曲线

- (1) $G(j\omega) = K$
- (2) $G(j\omega) = j\omega$
- (3) $G(j\omega) = 1/j\omega$
- (4) $G(j\omega) = 1/(1 + j\omega T)$
- (5) $G(j\omega) = 1 + j\omega T$
- (6) $G(j\omega) = 1/\left(1 - \frac{\omega^2}{\omega_n^2} + j2\xi \frac{\omega}{\omega_n}\right)$
- (7) $G(j\omega) = 1 - \frac{\omega^2}{\omega_n^2} + j2\xi \frac{\omega}{\omega_n}$



幅相频率特性

● 开环幅相特性曲线的绘制

$$G(s) = \frac{K(\tau_1 s + 1) \cdots (\tau_m s + 1)}{s^v (T_1 s + 1) \cdots (T_{n-v} s + 1)} = \frac{K \prod_{i=1}^m (\tau_i s + 1)}{s^v \prod_{j=1}^{n-v} (T_j s + 1)}$$

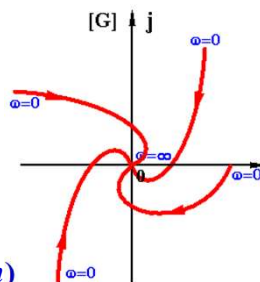
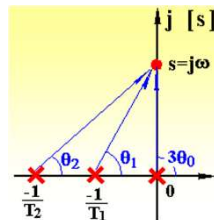
$$\left\{ \begin{aligned} |G(j\omega)| &= \frac{K \prod_{i=1}^m |1 + j\tau_i \omega|}{|\omega|^v \prod_{j=1}^{n-v} |1 + jT_j \omega|} = \frac{K \prod_{i=1}^m \sqrt{1 + \tau_i^2 \omega^2}}{\omega^v \prod_{j=1}^{n-v} \sqrt{1 + T_j^2 \omega^2}} \\ \varphi(\omega) &= \angle G(j\omega) = \sum_{i=1}^m \angle(1 + j\tau_i \omega) - v \times 90^\circ - \sum_{j=1}^{n-v} \angle(1 + jT_j \omega) \\ &= \arctan \tau_1 \omega + \cdots + \arctan \tau_m \omega \\ &\quad - v \times 90^\circ - \arctan T_1 \omega - \cdots - \arctan T_{n-v} \omega \end{aligned} \right.$$



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例: $G(s) = \frac{K}{s^v (T_1 s + 1)(T_2 s + 1)} = \frac{K/(T_1 T_2)}{s^v (s + 1/T_1)(s + 1/T_2)}$

v	$G(j\omega)$	$G(j0)$	$G(j\infty)$
0	$\frac{K}{(1 + j\omega T_1)(1 + j\omega T_2)}$	$K \angle 0^\circ$	$0 \angle -180^\circ$
I	$\frac{K}{j\omega(1 + j\omega T_1)(1 + j\omega T_2)}$	$\infty \angle -90^\circ$	$0 \angle -270^\circ$
II	$\frac{K}{(j\omega)^2(1 + j\omega T_1)(1 + j\omega T_2)}$	$\infty \angle -180^\circ$	$0 \angle -360^\circ$
III	$\frac{K}{(j\omega)^3(1 + j\omega T_1)(1 + j\omega T_2)}$	$\infty \angle -270^\circ$	$0 \angle -450^\circ$



起点 $\begin{cases} K \angle 0^\circ & v = 0 \\ \infty \angle -90^\circ v & v > 0 \end{cases}$ 终点 $0 \angle -90^\circ(n-m)$



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例: $G_1(s) = \frac{K}{s^2(T_1 s + 1)(T_2 s + 1)}$

$G_1(j0) = \infty \angle -180^\circ$

$\downarrow |G_1| \downarrow \angle G_1 \downarrow$

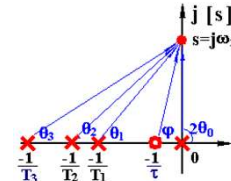
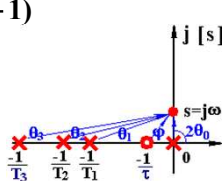
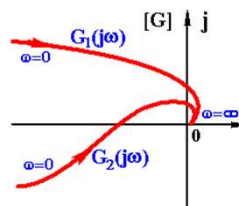
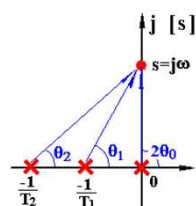
$G_1(j\infty) = 0 \angle -360^\circ$

$G_2(s) = \frac{K(\tau s + 1)}{s^2(T_1 s + 1)(T_2 s + 1)(T_3 s + 1)}$

$G_2(j0) = \infty \angle -180^\circ$

$\downarrow$

$G_2(j\infty) = 0 \angle -360^\circ$





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例: $G(s) = \frac{s^3}{(s+0.2)(s+1)(s+5)} = \frac{s^3}{(1+5s)(1+s)(1+0.2s)}$

$$G(j0) = 0 \angle 270^\circ$$

$$\downarrow |G| \uparrow \quad \angle G \downarrow$$

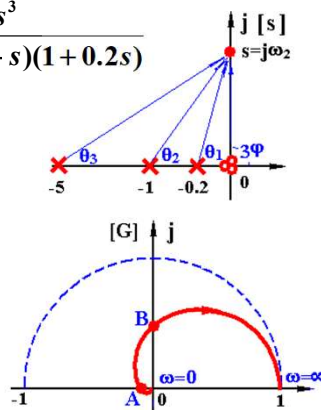
$$G(j\infty) = 1 \angle 0^\circ$$

$$G(j\omega) = \frac{-j\omega^3(1-j5\omega)(1-j\omega)(1-j0.2\omega)}{(1+25\omega^2)(1+\omega^2)(1+0.04\omega^2)}$$

$$= \frac{-\omega^4(6.2-\omega^2)-j\omega^3(1-6.2\omega^2)}{(1+25\omega^2)(1+\omega^2)(1+0.04\omega^2)}$$

$$= X + jY$$

$$A: \begin{cases} \omega_A = 1/\sqrt{6.2} = 0.402 \\ G(j\omega_A) = -0.0267 + j0 \end{cases} \quad B: \begin{cases} \omega_B = \sqrt{6.2} = 2.49 \\ G(j\omega_B) = 0 + j0.412 \end{cases}$$



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例: $G(s) = \frac{5}{s(s+1)(2s+1)}$

$$G(j\omega) = \frac{5}{j\omega(1+j\omega)(1+j2\omega)} = \frac{-j5(1-j\omega)(1-j2\omega)}{\omega(1+\omega^2)(1+4\omega^2)}$$

$$= \frac{-15}{(1+\omega^2)(1+4\omega^2)} - j \frac{5(1-2\omega^2)}{\omega(1+\omega^2)(1+4\omega^2)}$$

$$G(j0) = \infty \angle -90^\circ$$

$$G(j\infty) = 0 \angle -270^\circ$$

渐近线: $\text{Re}[G(j0)] \Rightarrow -15$

与实轴交点: $\text{Im}[G(j\omega)] = 0 \Rightarrow \omega = 1/\sqrt{2} = 0.707$

$$\text{Re}[G(j0.707)] = \frac{-15}{(1+0.5)(1+4 \times 0.5)} = -\frac{10}{3}$$

